I. Institution of Financial System

1. Finance

Problem: the separation between resources and talent/technology
(most of talented people do not have rich father, unfortunately!)

Nature: Finance is to combine resources and talents/technology together

Feature: Finance is about transferring or exchanging goods/values between people *across time* (macroeconomics: GDP=C+I+NX=C+S ➔ S=I+NX)

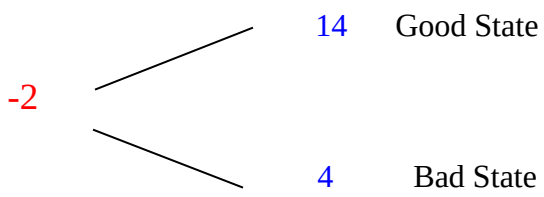
Frictions in the real word:
Contract enforcement
Information asymmetry (adverse selection and moral hazard)
Agency problems
…

Infrastructures for good finance:
Legal system
Accounting system
Valuation system (active secondary market)

Financial instruments:
Debt
Equity (information sensitive security)
…

Interaction between primary market and secondary market

Example: An entrepreneur has a technology/project with the payoffs of 14 in Good state and 4 in Bad state, which will be realized in one period. The project needs the initial investment, 2. The entrepreneur has no investment resources and has to rely on the outside financing.



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Three difficulties:

a. (Valuation) how to split the “pie” between the entrepreneur and outsider investors

The underlying question is

PV=?

14 Good State

4 Bad State

(time value, risk,….)

b. (Contract enforcement) how to guarantee that the entrepreneur will honor his promise to pay back to the investors

c. (Information disclosure) How to guarantee the entrepreneur will not lie, i.e., the entrepreneur can report the "Bad" state when the "Good" state is actually realized

Scenario 1: Equity financing (suppose outsider investors claim 50% shares)

Entrepreneur

7 Good State

2 Bad State

Investors

-2

7 Good State

2 Bad State

(Clearly, the entrepreneur and the investors are exchanging resource/values *across time*)

Incentives to lie for the entrepreneur in "Good" state

Lie by reporting "bad" state		Tell truth in "good" state	
(14-2)	-	7	= 5

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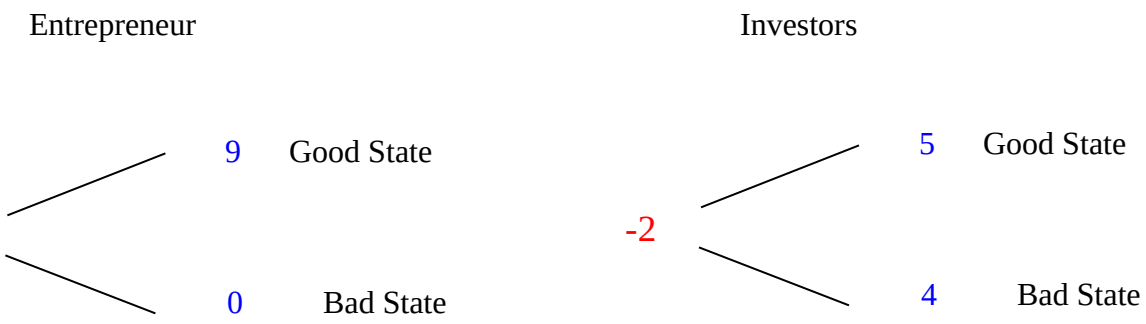
Scenario 2: Debt financing (suppose outsider investors claim face value as 5)



Incentives to lie for the entrepreneur in "Good" state

Lie by reporting "bad" state		tell truth in "good" state	
(14-4)	-	9	= 1

Message:
Equity is more information-sensitive than debt and thus demands more disclosures.

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2. Interest Rates

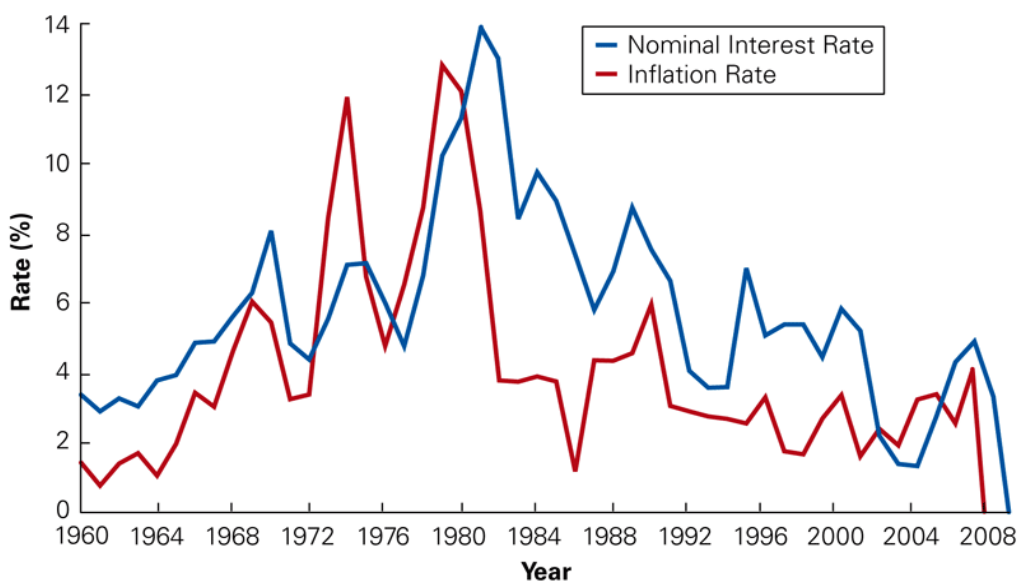
- Inflation and Real Versus Nominal Rates
  - **Nominal Interest Rate:** The rates quoted by financial institutions and used for discounting or compounding cash flows
  - **Real Interest Rate:** The rate of growth of your purchasing power, after adjusting for inflation

$$\text{Growth in Purchasing Power} = 1 + r_r = \frac{1 + r}{1 + i} = \frac{\text{Growth of Money}}{\text{Growth of Prices}}$$

- The Real Interest Rate

$$r_r = \frac{r - i}{1 + i} \approx r - i$$

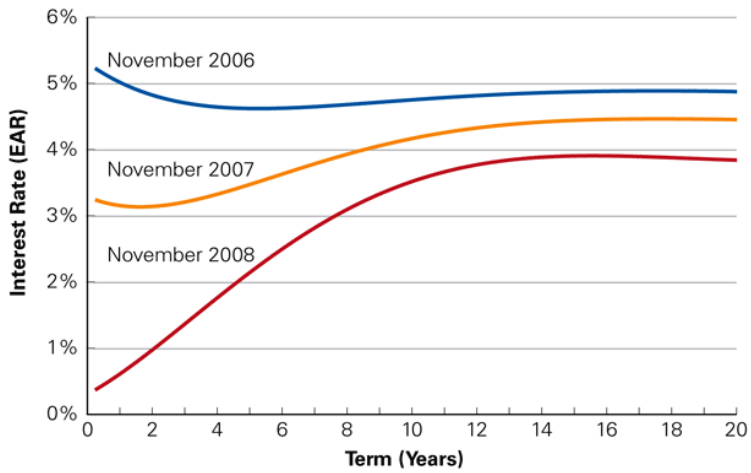
U.S. Interest Rates and Inflation Rates,1960–2009



- **Term Structure:** The relationship between the investment term and the interest rate.
- **Yield Curve:** A graph of the term structure.

Term Structure of Risk-Free U.S. Interest Rates, November 2006, 2007, and 2008

Term (years)	Date Nov-06	Nov-07	Nov-08
0.5	5.15%	3.20%	0.44%
1	5.02%	3.15%	0.60%
2	4.83%	3.14%	0.96%
3	4.71%	3.20%	1.35%
4	4.64%	3.32%	1.75%
5	4.62%	3.47%	2.13%
6	4.62%	3.63%	2.49%
7	4.65%	3.78%	2.81%
8	4.68%	3.93%	3.09%
9	4.71%	4.06%	3.32%
10	4.75%	4.17%	3.51%
15	4.87%	4.44%	3.90%
20	4.88%	4.45%	3.84%



- **Interest Determination**
  - The Federal Reserve determines very short-term interest rates through its influence on the **federal funds rate**, which is the rate at which banks can borrow cash reserves on an overnight basis.
  - All other interest rates on the yield curve are set in the market and are adjusted until the supply of lending matches the demand for borrowing at each loan term.
- **Interest Rate Expectations**
  - The shape of the yield curve is influenced by interest rate expectations. **Market expectations (pure expectations) hypothesis**

$$(1+i_n)^n = (1+i_1)(1+\mathbb{E}(f_{1,2}))(1+\mathbb{E}(f_{2,3}))\ldots(1+\mathbb{E}(f_{n-1,n}))$$

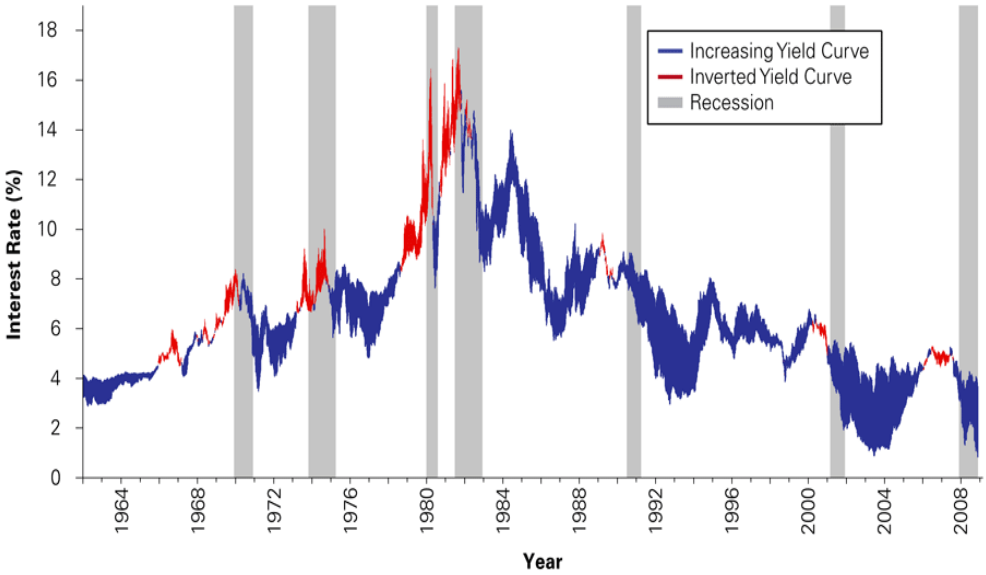
where i_t is the (annualized) spot rate for a t-years long-term loan , $f_{t-1,t}$ is the forward rate between date t-1 and date t, and E(.) is the expectation operator.

Liquidity premium theory

$$(1+i_n)^n = (1+i_1)(1+\mathbb{E}(f_{1,2}))(1+\mathbb{E}(f_{2,3}))\ldots(1+\mathbb{E}(f_{n-1,n})) + LP$$

Where *LP* is the liquidity premium associated with an *n* year bond.

- An inverted yield curve indicates that interest rates are expected to decline in the future.
    - Because interest rates tend to fall in response to an economic slowdown, an inverted yield curve is often interpreted as a negative forecast for economic growth.
        - Each of the last six recessions in the United States was preceded by a period in which the yield curve was inverted.
        - The yield curve tends to be sharply increasing as the economy comes out of a recession and interest rates are expected to rise.



Example: consider two periods

$$(1+i_1)(1+\mathbb{E}(f_{1,2}))+LP \quad = \quad (1+i_2)^2$$

$i_1 < i_2$ does not imply $i_1 < E(f_{1,2})$:

normal (upward) term structure does not predict a good economy ahead

$i_1 > i_2$ does imply $i_1 > E(f_{1,2})$:

inverted (downward) term structure does predict a bad economy ahead

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3. **Duration**

There is an inverse relationship between interest rates and bond prices.

- As interest rates and bond yields rise, bond prices fall.
- As interest rates and bond yields fall, bond prices rise.

We want to quantify how a bond's price changes with interest rate.

- **Duration**

  The sensitivity of a bond's price to changes in interest rates is measured by the bond's duration.

  - Bonds with high durations are highly sensitive to interest rate changes.
  - Bonds with low durations are less sensitive to interest rate changes.

  Example:

The Interest Rate Sensitivity of Bonds

Problem

Consider a 15-year zero-coupon bond and a 30-year coupon bond with 10% annual coupons. By what percentage will the price of each bond change if its yield to maturity increases from 5% to 6%?

Solution

First, we compute the price of each bond for each yield to maturity:

Yield to Maturity	15-Year, Zero-Coupon Bond	30-Year, 10% Annual Coupon Bond
5%	$\frac{100}{1.05^{15}} = \$48.10$	$10 \times \frac{1}{0.05}\left(1 - \frac{1}{1.05^{30}}\right) + \frac{100}{1.05^{30}} = \$176.86$
6%	$\frac{100}{1.06^{15}} = \$41.73$	$10 \times \frac{1}{0.06}\left(1 - \frac{1}{1.06^{30}}\right) + \frac{100}{1.06^{30}} = \$155.06$

The price of the 15-year zero-coupon bond changes by (41.73 − 48.10)/48.10 = −13.2% if its yield to maturity increases from 5% to 6%. For the 30-year bond with 10% annual coupons, the price change is (155.06 − 176.86)/176.86 = −12.3%. Even though the 30-year bond has a longer maturity, because of its high coupon rate its sensitivity to a change in yield is actually less than that of the 15-year zero coupon bond.

– Consider the case of continuously-compounded yields

–

$$V = \sum_{i=1}^{n} PV_i = \sum_{i=1}^{n} CF_i \cdot e^{-y \cdot t_i}$$

where:

- i indexes the cash flows,
- PV_i is the present value of the ith cash payment from an asset,
- CF_i is the cash flow of the ith payment from an asset,
- y is the yield to maturity (continuously-compounded) for the an asset,
- t_i is the time in years until the ith payment will be received,
- V is the present value of all cash payments from the asset until maturity.

The **duration** is a price sensitivity measure, defined as the percentage derivative of price with respect to yield, that is,

$$D = -\frac{1}{V}\frac{\partial V}{\partial y}$$

We prove

$$D = -\frac{1}{V}\frac{\partial V}{\partial y} = \sum_{i=1}^{n} \frac{PV_i}{V} \cdot t_i$$

Proof:

$$\begin{aligned} \frac{\partial V}{\partial y} &= \partial \left(\sum_{i=1}^{n} CF_i \cdot e^{-y \cdot t_i} \right) / \partial y \\ &= \sum_{i=1}^{n} CF_i \cdot e^{-y \cdot t_i} \left(-t_i\right) \\ &= \sum_{i=1}^{n} PV_i \cdot \left(-t_i\right) \end{aligned},$$

Thus, the duration (D) is calculated as

$$D = -\frac{1}{V}\frac{\partial V}{\partial y} = \sum_{i=1}^{n} \frac{PV_i}{V} \cdot t_i.$$

Intuitively, duration is the weighted average maturity of the present values of the cash flow.

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4. Information, competition and stock prices

- **Efficient Markets Hypothesis**
  - Implies that securities will be fairly priced, based on their future cash flows, given all information that is available to investors.

- **The Efficient Markets Hypothesis Versus No Arbitrage**

  - The efficient markets hypothesis states that securities with *equivalent risk* should have the *same expected return*.

- **Three forms of market efficiency**
  - **Weak form efficiency**

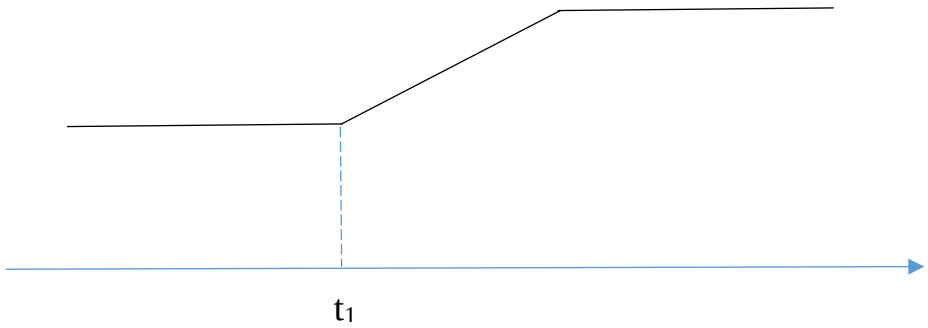
    It is impossible to consistently profit by trading on information in past prices (i.e. No price pattern). E.g., Technical analysis does not work.
  - **Semi-strong form efficiency**

    It is impossible to consistently profit by trading on any public information, such as news announcement – there should be a one-shot change in stock price at news announcement. E.g., Financial statement analysis does not work.
  - **Strong form efficiency**

    It should not be possible to consistently profit even by trading on private information.

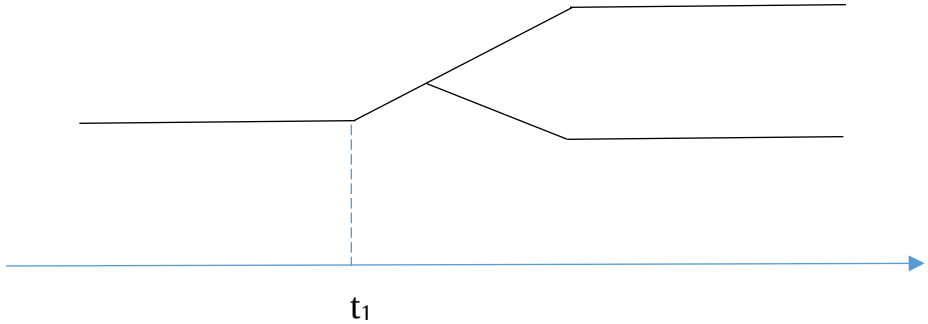
Example: t_0 is the board meeting date and t_1 is the earning announcement date

Not efficient at all:



(Trend will continue; not too late to buy and profit)

Weak form efficiency:



(Trend will either continue (momentum) or reverse; cannot profit based on pure technical analysis by finding price patterns. However, by using technical analysis combined with accounting (public) information analysis, it is possible to predict whether momentum or reversal will happen, and thus profit)

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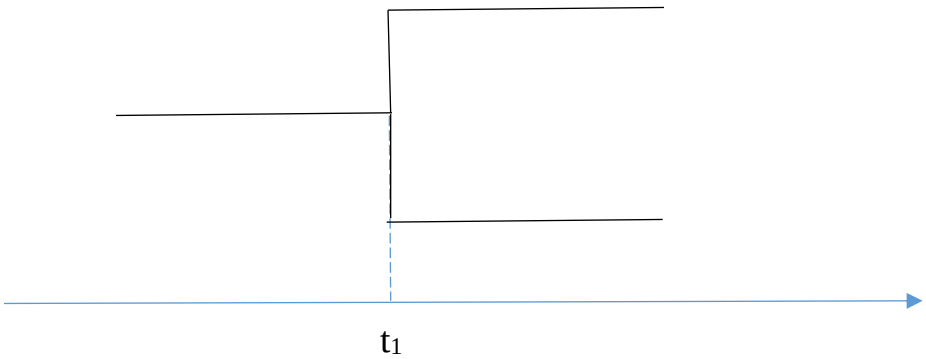
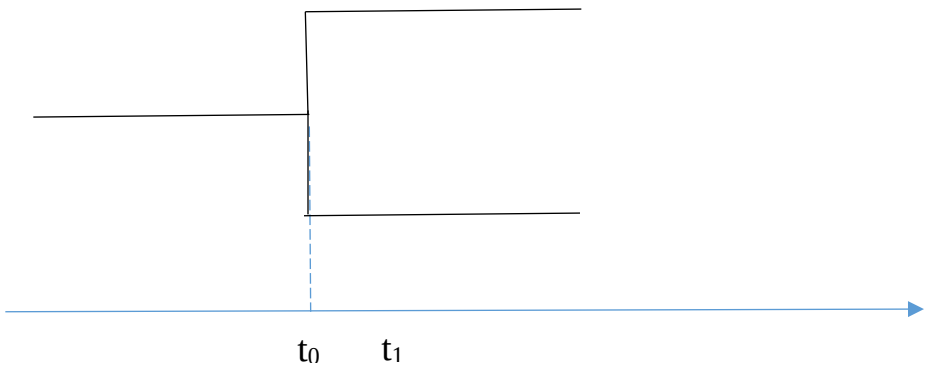
Semi-strong form efficiency:



(No trend at all; the stock price has a one-shot change -- jump -- when public news comes out)

Strong form efficiency:



(The one-shot change -- jump – occurs as soon as the inside information is available)

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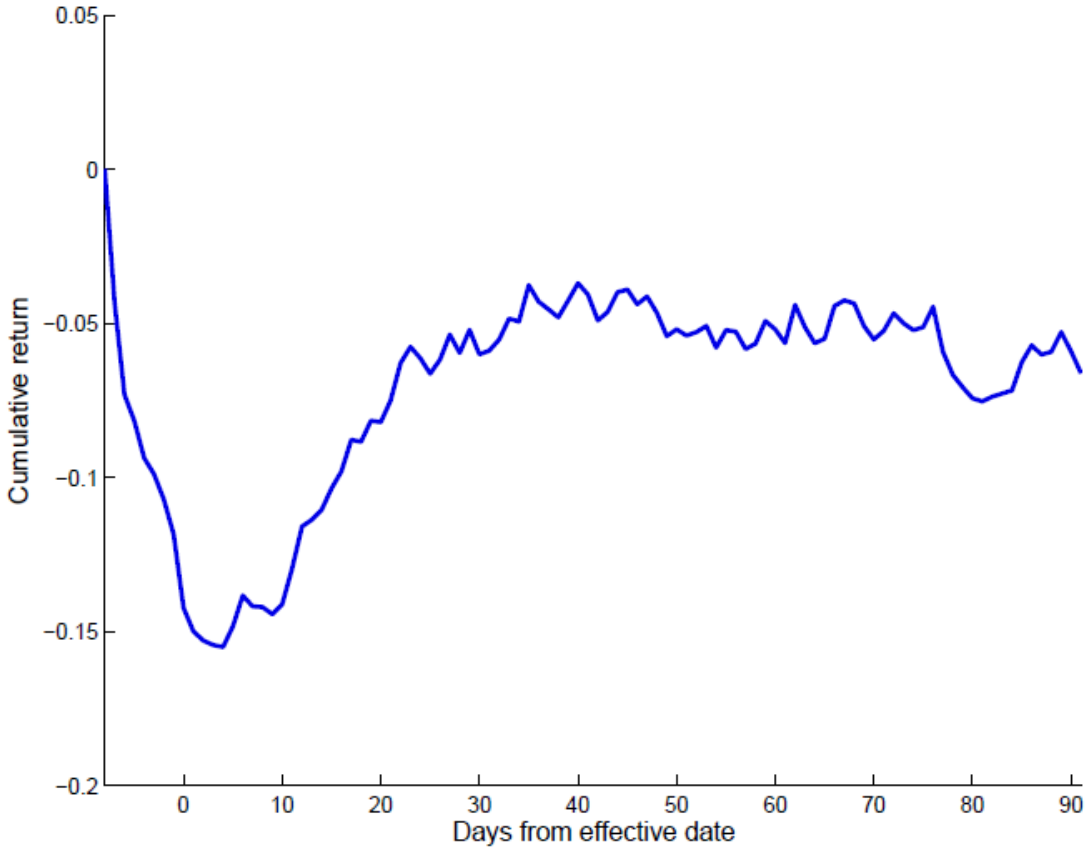
- **Some evidence on market efficiency**

Example 1: Index Deletions

The following figure illustrates the average price impact of deletions of equities from the S&P 500 stock index, and the associated average degree of price reversal.

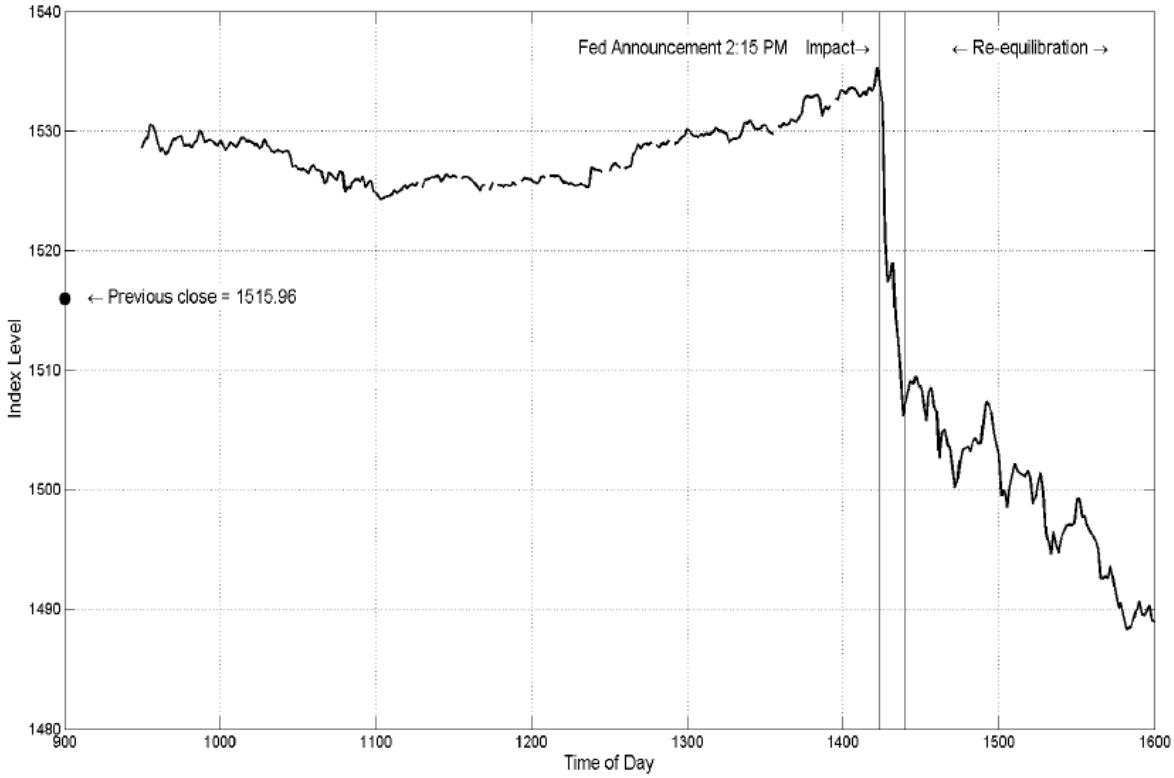


See Duffie (2010):
http://www.stanford.edu/~duffie/PresidentialAddressApril15NormalFormat.pdf

There exists a price pattern which investors can profit from. So even in the US market, sometimes market is not efficient at all.

Example 2: The Impact of the Federal Reserve's Interest Rate Target Announcement on Stock Prices

Intraday Behavior of the Forward Value of the S&P Index on Dec. 11, 2007



See Birru and Figlewski (2010):

http://pages.stern.nyu.edu/~sfiglews/Docs/Fed%20Announcements_MAR292010.pdf

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